

Final Project Report: TCP Modeling in Glioblastoma (CFB-GBM)

TCP Modeling GBM Project Group

2026-06-28

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1 Final Project Report: Tumor Control Probability Modeling in Glioblastoma

Course: TCP Modeling in GBM — Final Project

Dataset: CFB-GBM (TCIA Version 3, DOI 10.7937/v9pn-2f72)

Institution: Centre François Baclesse (open cohort via The Cancer Imaging Archive)

Modeling cohort: n = 190 (after DVH quality control)

RANO imaging subset: n = 137 (t0 to t1 labels)

Report date: 2026-06-28

1.1 Abstract

We implemented a reproducible Python pipeline for Tumor Control Probability (TCP) modeling on the open-access CFB-GBM glioblastoma radiotherapy cohort. The workflow covers clinical data curation, NIfTI-based DVH feature extraction, four TCP model families (Poisson, logistic, probit, gEUD-based), bootstrap confidence intervals, survival analysis, and Version 3 RANO imaging endpoints with PyRadiomics comparison.

Pooled EQD2 discriminated an exploratory overall survival (OS) median-split proxy (ROC AUC approximately 0.68) but not RANO non-progressive disease when dose was the sole predictor (AUC approximately 0.43). After adjusting for fractionation scheme and clinical covariates, pre-treatment GTV volume and author-provided PyRadiomics features predicted early RANO with moderate cross-validated discrimination (pooled LOOCV AUC approximately 0.64; hypofractionated arm LOOCV approximately 0.74; PyRadiomics nested CV approximately 0.74).

Conclusion: classical pooled TCP dose-response validation is not supported on routine CFB-GBM data because of limited within-protocol dose spread and endpoint confounding. The project delivers

a complete open pipeline, formal negative result, and tumour-burden benchmarks for future TCP feasibility studies.

Keywords: glioblastoma; tumor control probability; DVH; EQD2; RANO; PyRadiomics; CFB-GBM; open data

1.2 1. Introduction and Clinical Background

1.2.1 1.1 Glioblastoma and standard radiotherapy

Glioblastoma (GBM) is the most common malignant primary brain tumour in adults. Maximal safe resection followed by radiotherapy with concurrent temozolomide remains the standard of care for fit patients (Stupp protocol: 60 Gy delivered in 30 fractions of 2 Gy) [1]. Despite a uniform nominal protocol, observed outcomes vary widely because of age, performance status, molecular factors, and tumour burden at baseline [2].

In elderly or frail patients, shorter hypofractionated schedules are widely used. The CFB-GBM cohort includes 40.05 Gy in 15 fractions, consistent with pragmatic GBM management described in Nordic and short-course trials [3,15]. These two fractionation schemes differ not only in total dose and fraction size but also in patient selection: in our cohort, 60 Gy patients were younger (median age 65 vs 75 years) and had longer OS (median 60 vs 28 weeks).

1.2.2 1.2 Tumor Control Probability (TCP) modeling

TCP models estimate the probability that all clonogenic tumour cells are sterilised after radiotherapy, as a function of delivered dose and radiobiological parameters such as D50 (dose for 50% control probability) and gamma50 (steepness of the dose-response curve) [4,5]. TCP models are used in treatment-planning research, protocol comparison, and hypothetical dose escalation studies.

Valid TCP validation requires three conditions emphasized in the radiotherapy modeling literature [7]:

1. **A tumour-control endpoint** (local control or a validated imaging surrogate), not a distant-survival proxy alone.
2. **Meaningful inter-patient dose heterogeneity** within the structure of interest (typically GTV Dmean spread greater than approximately 1 Gy in TCP audit workflows).
3. **Adequate sample size** and pre-specified cross-validation when predictive models are reported.

1.2.3 1.3 CFB-GBM and project aims

The CFB-GBM dataset (Centre François Baclesse, $n = 264$ in the public TSV release) provides pre- and post-treatment MRI, RTDOSE maps, GTV segmentations, clinical tables, and—since Version 3—RANO response labels and PyRadiomics feature tables [8–10]. The dataset is distributed as pre-processed NIfTI rather than full DICOM-RT plans.

Primary aim: build a reproducible TCP modeling pipeline on this open cohort, estimate D50 and gamma50 with uncertainty, compare models, and relate our parameters to published TCP literature.

Secondary aim (Version 3 extension): determine whether early RANO imaging response and PyRadiomics features rescue dose-response signal when classical pooled TCP fails, and report honest out-of-sample performance (LOOCV and nested cross-validation).

1.3 2. Dataset Description and Cohort Construction

1.3.1 2.1 Source data and imaging format

Item	Value
Data repository	The Cancer Imaging Archive (TCIA)
Collection	CFB-GBM Version 3
DOI	10.7937/v9pn-2f72
Total patients in clinical TSV	264
Patients with RTDOSE flag	194
Patients with GTV segmentation	191
Included for TCP modeling	190
Primary exclusion	Unknown RT dose (70 patients)
DVH QC exclusion	1 (patient 32, GTV Dmean = 0 Gy)
Fractionation scheme A	120 patients: 60 Gy / 30 fractions
Fractionation scheme B	61 patients: 40.05 Gy / 15 fractions
Median age (modeling cohort)	70 years
Median overall survival	51 weeks
Sex (M / F)	117 / 73
Median GTV volume	36.2 cc
Median GTV Dmean	59.84 Gy
RANO t0 to t1 labels available	137 / 190 (72%)
RANO labels in 40 Gy arm	34 patients

Structure set note: all dosimetric metrics refer to **GTV at t0** only. CTV and PTV contours are not included in the public CFB-GBM release (GTV-only segmentation). RTPLAN and RTSTRUCT DICOM objects are not distributed on TCIA Version 3; the cohort provides co-registered **RTDOSE** and **GTV Nifti** volumes. D95, D98, D50, D2, gEUD, EQD2, and GTV volume were computed from voxel dose grids and binary masks—the same quantities typically exported from a treatment-planning system after structure delineation and dose calculation.

1.3.2 2.2 Cohort flow (264 to 190)

Starting from 264 patients in the clinical table, 70 were excluded because prescribed RT dose was unknown or missing. Among imaging-eligible patients, one patient failed DVH quality control (zero dose to GTV on RTDOSE). The final modeling table contains 190 rows and 58 columns (`modeling_table.csv`), including clinical covariates, DVH scalars, EQD2, RANO categories, and PyRadiomics merge keys. Notebook 01 (`01_cohort_overview.ipynb`) documents the full exclusion breakdown and demographic tables.

1.3.3 2.3 Outcome definitions

Primary TCP endpoint (exploratory): OS greater than or equal to cohort median (51 weeks), used as a binary outcome for dose–response modeling. This is not equivalent to formal tumour control.

Secondary endpoint (Version 3): RANO non-progressive disease at t1, defined as stable disease, minor response, partial response, or complete response versus progressive disease [12,13]. RANO non-PD rate in the modeling subset was approximately 77%.

1.4 3. Data Acquisition, Curation, and Reproducibility

1.4.1 3.1 Pipeline overview

1. Download clinical, treatment, RANO, and PyRadiomics TSV files (`python -m src.data.download_clinical`).
2. Merge tables and apply inclusion rules (`src/data/cohort_builder.py`); export `cohort.csv`.
3. Download RTDOSE and GTV t0 NIfTI for 191 patients (IBM Aspera Connect; approximately 52 GB local storage).
4. Verify file completeness (`verify_raw_data.py`).
5. Compute cumulative DVH and scalar metrics (`feature_builder.py`, `dvh_calculator.py`).
6. Apply DVH QC (`dvh_qc.py`); export `modeling_table.csv`.
7. Fit TCP models, survival models, and RANO prediction suite; export summary tables and figures.

All analysis code lives in `src/`. Random seed 42 is fixed for bootstrap and cross-validation splits. Notebooks 01–06 reproduce the analysis end-to-end after `pip install -r requirements.txt`.

1.4.2 3.2 Imaging format and DVH inputs

Typical DICOM-RT workflow	CFB-GBM Version 3	This study
RTPLAN / RTSTRUCT	Not in public download	RTDOSE + GTV NIfTI
CTV / PTV structures	Not released	GTV t0 only

Typical DICOM-RT workflow	CFB-GBM Version 3	This study
RTDOSE + structure	RTDOSE + GTV NIfTI	<code>dvh_calculator.py</code> , <code>feature_builder.py</code>
Scalar DVH metrics	D95, D98, D50, D2, volume, gEUD, EQD2	Exported to <code>modeling_table.csv</code>

Given RTDOSE and GTV masks, cumulative DVH and standard metrics match those derived from a DICOM plan export.

1.4.3 3.3 EQD2 correction

To compare the two fractionation schemes on a common biologically effective scale, we computed equivalent dose in 2 Gy fractions (EQD2) using the linear-quadratic model with $\alpha/\beta = 10$ Gy for GTV [6,11]:

$$\text{EQD2} = D_{\text{total}} \cdot \frac{d_{\text{fx}} + \alpha/\beta}{2 + \alpha/\beta}$$

1.5 4. Dosimetric Feature Extraction

1.5.1 4.1 DVH computation

For each patient, cumulative dose-volume histograms (DVH) were computed inside the t0 GTV mask from registered RTDOSE grids. Scalar metrics exported per patient include:

Metric	Description
Dmean	Mean dose to GTV (Gy)
D95, D98, D50, D2	Dose received by at least x% of GTV volume
Dmax	Maximum dose to GTV
volume_cc	GTV volume (cm ³)
Vx	Percentage of GTV receiving at least x Gy
gEUD_a	Generalised equivalent uniform dose for a = -10, 1, 10
HI	Homogeneity index inside GTV
eqd2_gy	EQD2 as defined above

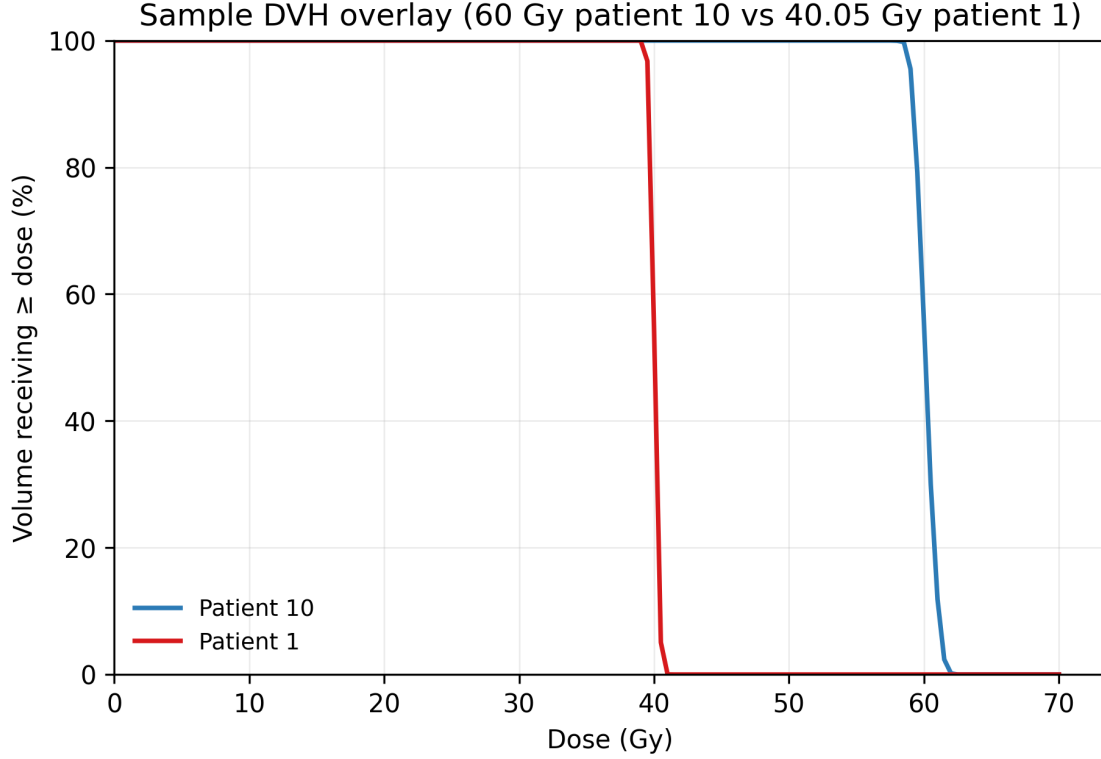


Figure 1: Sample cumulative DVH curves for three patients (GTV t0, RTDOSE).

Figure (DVH). Representative cumulative dose–volume histograms inside the t0 GTV mask, illustrating inter-patient dose distribution at fixed protocol dose (60 Gy arm) versus hypofractionation (40 Gy arm). Full cohort mean DVH: [figures/02_dvh_cohort_mean.png](#).

1.5.2 4.2 Dose heterogeneity audit

A central feasibility finding is that **within-arm GTV Dmean variation is negligible**. In the 60 Gy arm, GTV Dmean standard deviation was only **0.28 Gy**. Therefore, classical dose-response fitting *within* a single protocol arm lacks statistical power regardless of TCP model choice [7]. Between-arm dose differences reflect different fractionation schedules and patient selection, not planned dose escalation.

1.6 5. Mathematical Description of TCP Models

We implemented four binary-outcome TCP formulations following standard radiobiology references [4–7]:

Model	Parameters	Formula (conceptual)
Poisson TCP	D50, gamma50	Clonogenic survival with exponential shoulder

Model	Parameters	Formula (conceptual)
Logistic TCP	D50, k	Sigmoid dose-response
Probit TCP	D50, sigma	Normal cumulative distribution link
gEUD TCP	D50, gamma50, a	DVH-collapsed dose metric then Poisson-style TCP

Full equations are documented in `reports/manuscript_equations_fragment.tex`. Parameters were estimated by **maximum likelihood** using `scipy.optimize.minimize` with method **L-BFGS-B** and box constraints on D50 and gamma50 (`src/models/poisson_tcp.py`). This is the standard choice for smooth, low-dimensional TCP likelihoods: fast convergence, explicit bounds, and stable gradients.

1.6.1 5.1 Maximum-likelihood optimization

Poisson TCP parameters were estimated by constrained maximum likelihood (`scipy.optimize.minimize`, method L-BFGS-B). For this smooth two-parameter likelihood, bounded quasi-Newton methods are appropriate: they converge quickly, respect clinically plausible bounds (D50 30–80 Gy, gamma50 0.5–8), and remain stable across bootstrap resamples.

We cross-checked the MLE with TNC, SLSQP, Powell, Nelder-Mead, and differential evolution on the full cohort ($n = 190$). All methods recovered $D50 = 53.20$ Gy within 0.002 Gy; Nelder-Mead and differential evolution required $2\text{--}20\times$ more computation time without improving the fit.

Model comparison used AIC, BIC, ROC AUC, Brier score, and Hosmer-Lemeshow calibration.

1.6.2 5.2 Uncertainty quantification (bootstrap and profile likelihood)

Bootstrap 95% confidence intervals (1000 resamples, seed 42) and profile-likelihood 95% intervals were computed for Poisson D50 and gamma50 [18]. Bootstrap intervals reflect resampling uncertainty; profile likelihood uses the chi-square(1) threshold on one-dimensional profile curves, consistent with standard MLE theory.

1.7 6. Parameter Estimation and Four-Model Comparison

1.7.1 6.1 Pooled TCP on OS proxy (EQD2, $n = 190$)

Model	k	AIC	BIC	ROC AUC	Brier	HL p-value
probit_tcp	2	241.73	248.23	0.684	0.218	0.101
logistic_tcp	2	241.75	248.24	0.684	0.218	0.101
poisson_tcp	2	241.82	248.31	0.684	0.218	0.100
eud_tcp (gEUD)	3	243.69	253.43	0.685	0.218	0.784

Model	k	AIC	BIC	ROC AUC	Brier	HL p-value
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All three two-parameter models are nearly identical on this cohort. The Poisson fit is reported for literature comparison.

1.7.2 6.2 Bootstrap confidence intervals (Poisson, EQD2)

Parameter	Estimate (Gy)	95% Bootstrap CI	Bootstrap SD
D50	53.20	[49.54, 56.75]	1.87
gamma50	3.32	[2.06, 4.69]	0.67

1.7.3 6.3 Profile-likelihood confidence intervals (Poisson, EQD2)

Parameter	Estimate	95% Profile-LI CI
D50 (Gy)	53.20	[48.91, 58.15]
gamma50	3.32	[1.76, 4.85]

Profile and bootstrap intervals are similar in width; both exclude uninformative values (gamma50 > 1).

Likelihood-ratio test vs null model: p approximately 3×10^{-6} . Five-fold cross-validation AUC: 0.68 +/- 0.10.

1.7.4 6.4 Calibration — observed vs predicted

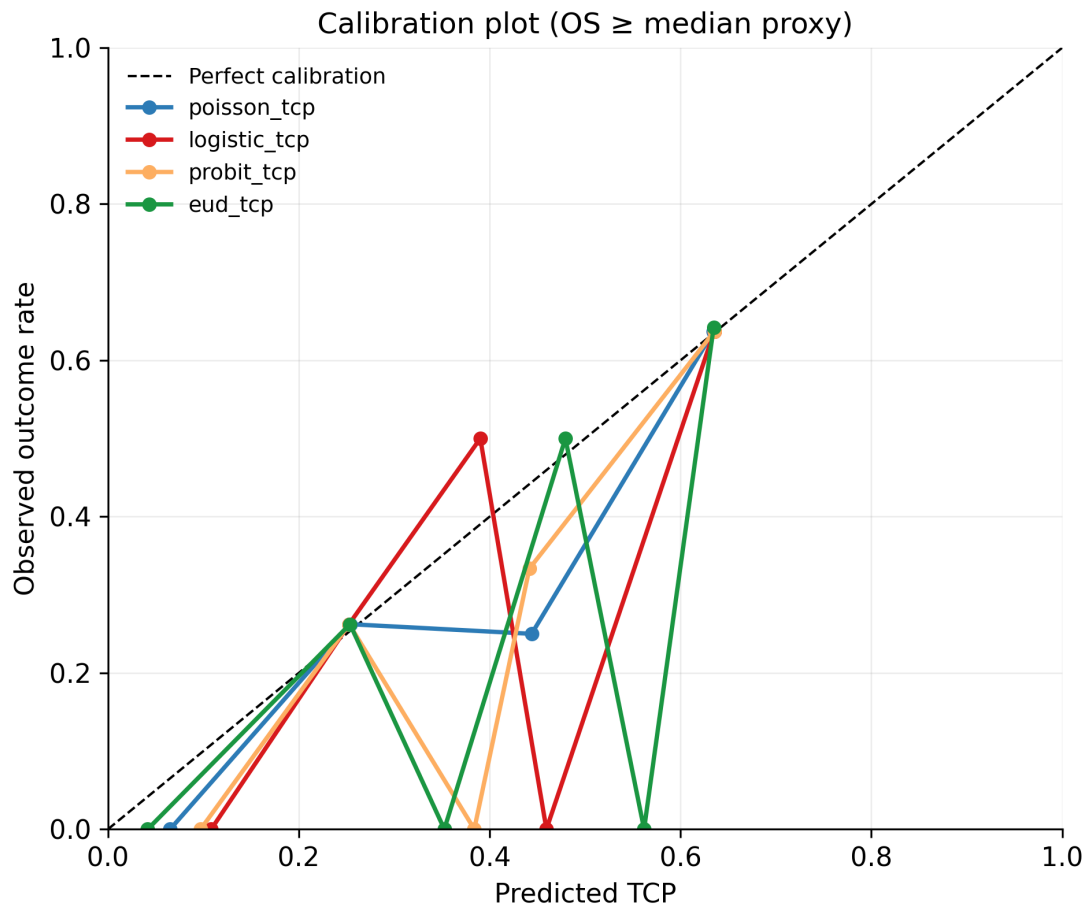


Figure 2: Calibration plot: predicted TCP (Poisson, EQD2) vs observed OS proxy rate by decile.

Figure (calibration). Hosmer-Lemeshow groups (10 bins): observed fraction with OS $\geq$ median vs mean predicted TCP per bin. HL p approximately 0.10; no systematic over- or under-prediction across dose deciles.

1.8 7. Results

1.8.1 7.1 Clinical prognosis and confounding

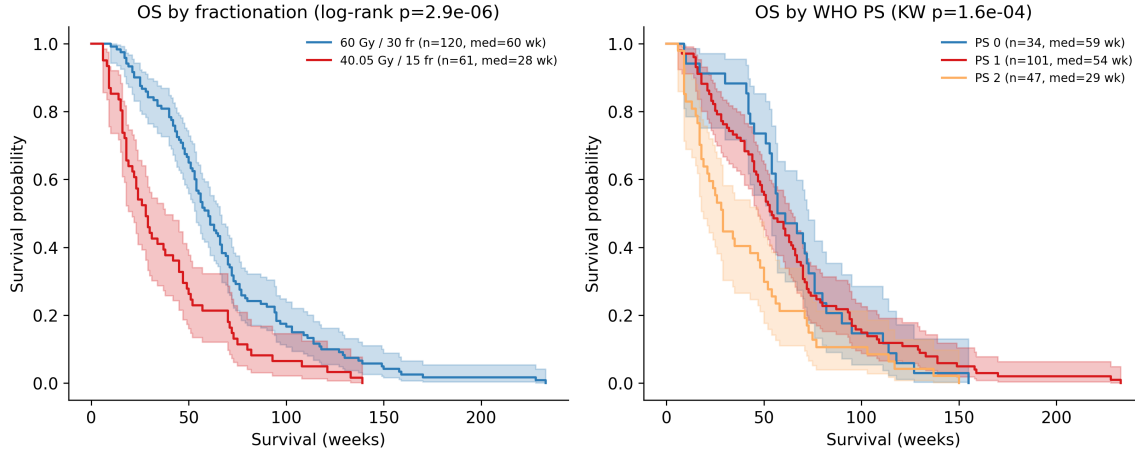


Figure 3: Figure 1. Kaplan-Meier overall survival by fractionation scheme and WHO performance status.

Figure 1. Kaplan-Meier OS by fractionation scheme and WHO performance status ($n = 190$). Sixty Gy patients had median OS 60 weeks vs 28 weeks for 40 Gy (log-rank p approximately 3×10^{-6}). Cox model (age + sex + WHO PS + scheme): scheme HR approximately 0.54 (p approximately 0.0007), WHO PS HR approximately 1.42 (p approximately 0.001).

OS by WHO PS	n	Median OS (weeks)	IQR (weeks)
PS 0	34	59	46–79
PS 1	101	54	32–77
PS 2	47	29	17–55

Kruskal-Wallis across PS groups: p approximately 1.6×10^{-4} . Confounding audit: Pearson $r(\text{age, EQD2}) = -0.57$; older patients disproportionately received hypofractionation.

1.8.2 7.2 Pooled TCP — OS proxy

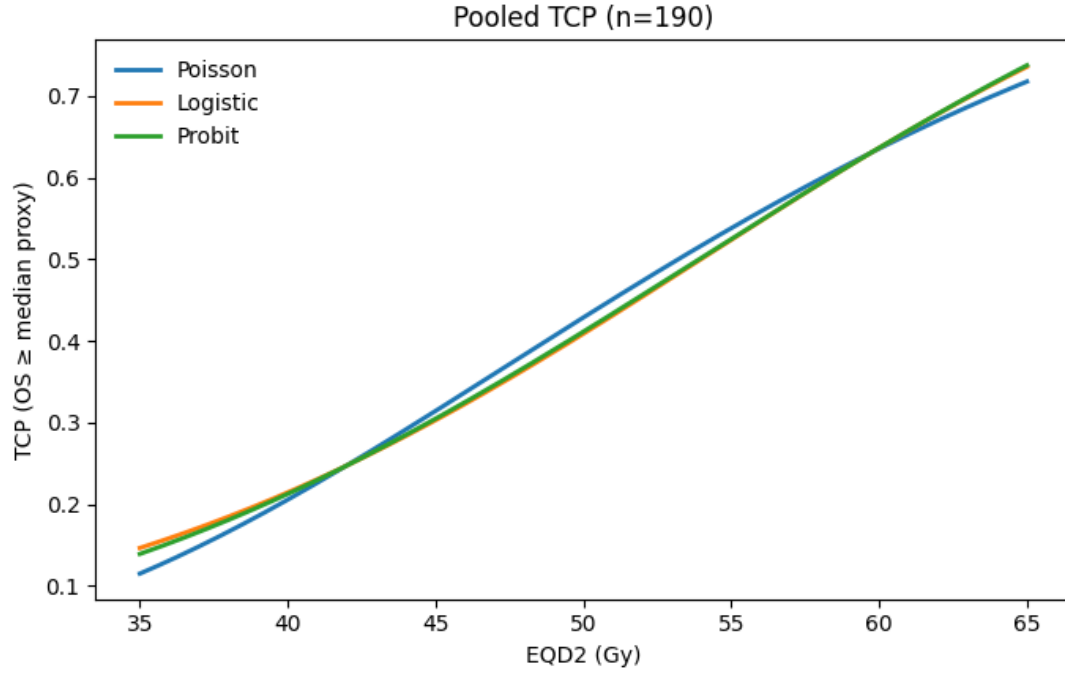


Figure 4: Figure 2. TCP dose-response curves for EQD2 vs OS median-split proxy.

Figure 2. Fitted Poisson, logistic, probit, and gEUD TCP curves (EQD2 vs OS $\geq$ median, $n = 190$). In-sample AUC approximately 0.68. Interpretation: association partly reflects fractionation and patient selection, not within-protocol dose escalation.

1.8.3 7.3 Pooled TCP — RANO endpoint failure

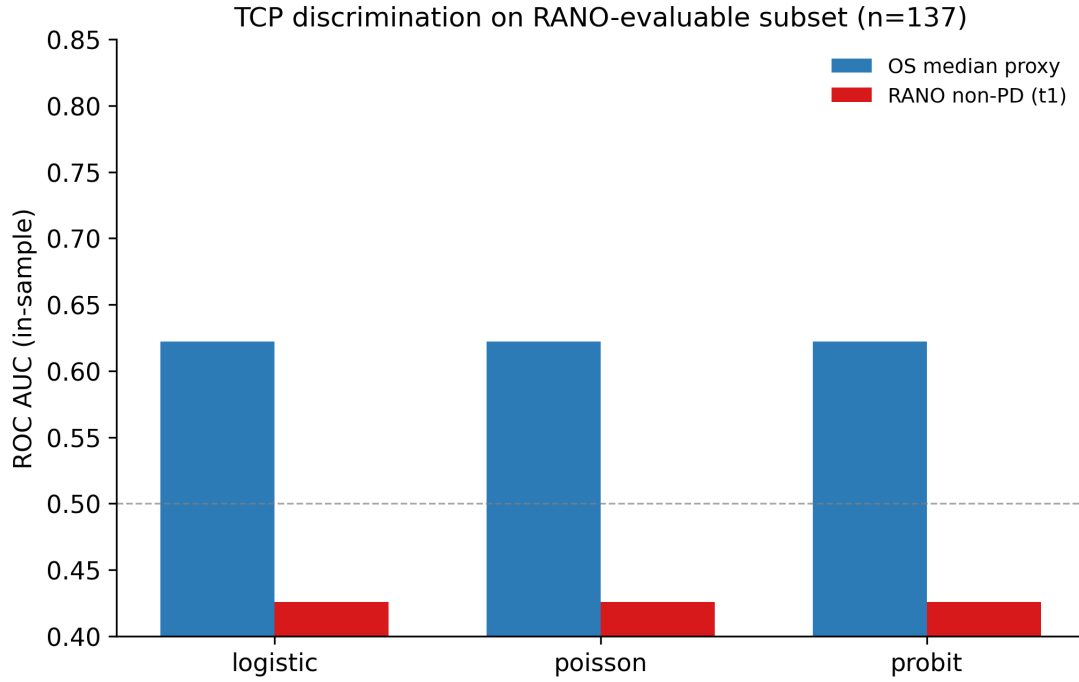


Figure 5: Figure 3. OS vs RANO TCP endpoint comparison on the same patients.

Figure 3. Poisson TCP (EQD2) on 137 RANO-labelled patients: OS proxy AUC approximately 0.62; **RANO non-PD AUC approximately 0.43** (LR $p = 1.0$ vs null). Higher EQD2 correlated with more early PD at t1 (27% PD in 60 Gy arm vs 15% in 40 Gy arm) despite better long-term OS.

Endpoint	n	Poisson AUC	5-fold CV AUC
OS median-split	137	0.622	0.621
RANO non-PD	137	0.426	0.422

1.8.4 7.4 Within-arm stratified analysis

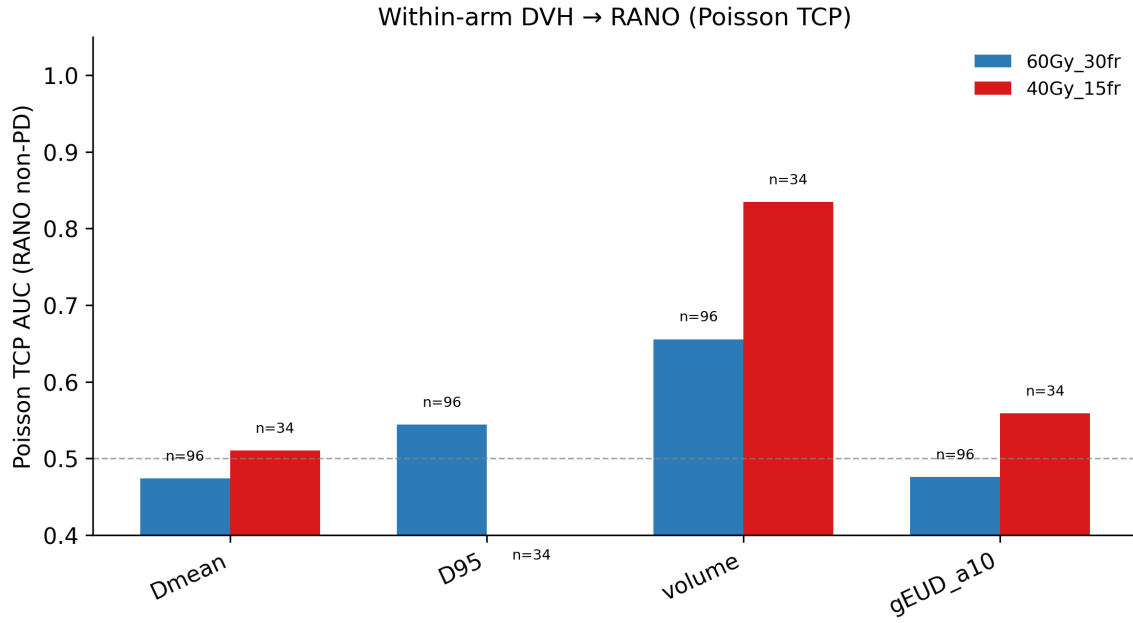


Figure 6: Figure 4. Within-arm DVH metrics predicting RANO non-PD.

Figure 4. Within-arm Spearman and Poisson TCP AUC by fractionation scheme. Sixty Gy arm ($n = 96$): GTV Dmean SD = 0.28 Gy; no dose-response. Forty Gy arm ($n = 34$): GTV volume predicted RANO (Spearman rho approximately 0.41, p approximately 0.016; Poisson AUC = 0.83, LR p approximately 0.037).

1.8.5 7.5 Hypofractionated arm — multivariable logistic regression

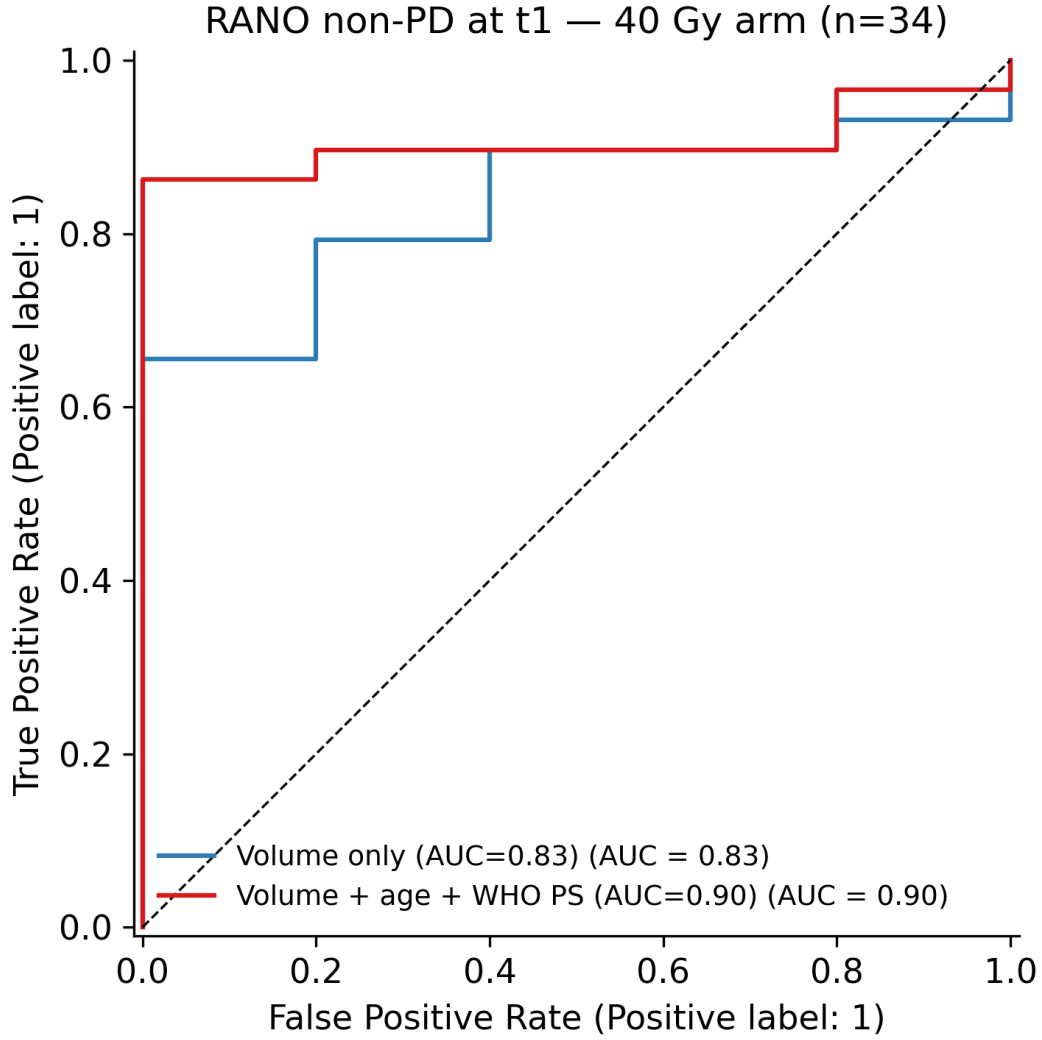


Figure 7: Figure 5. ROC curves for 40 Gy arm RANO models.

Figure 5. Multivariable logistic regression in the 40.05 Gy arm ($n = 34$): volume-only AUC = 0.83; volume + age + WHO PS AUC = 0.90 in-sample.

Model (40 Gy arm)	n	In-sample AUC	LOOCV AUC
volume only	34	0.834	0.745
volume + age + PS	34	0.903	0.697

LOOCV confirms signal beyond chance but reduces optimism relative to in-sample AUC = 0.90.

1.8.6 7.6 Volume validation

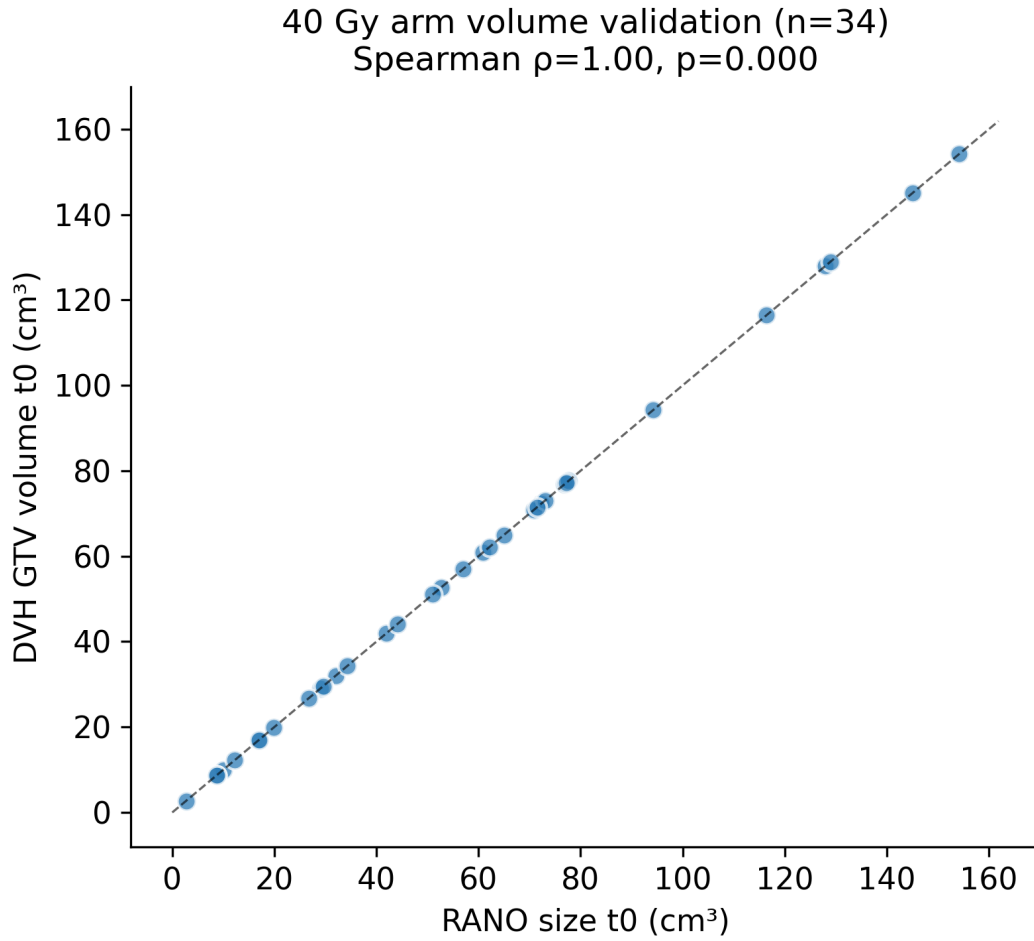


Figure 8: Figure 6. DVH GTV volume vs author RANO t0 volume.

Figure 6. Agreement between DVH-derived GTV volume and author-reported RANO size_t0_cm3 (Spearman rho = 1.00, n = 141). Validates the feature extraction pipeline.

1.8.7 7.7 Pooled RANO models (volume + clinical + scheme)

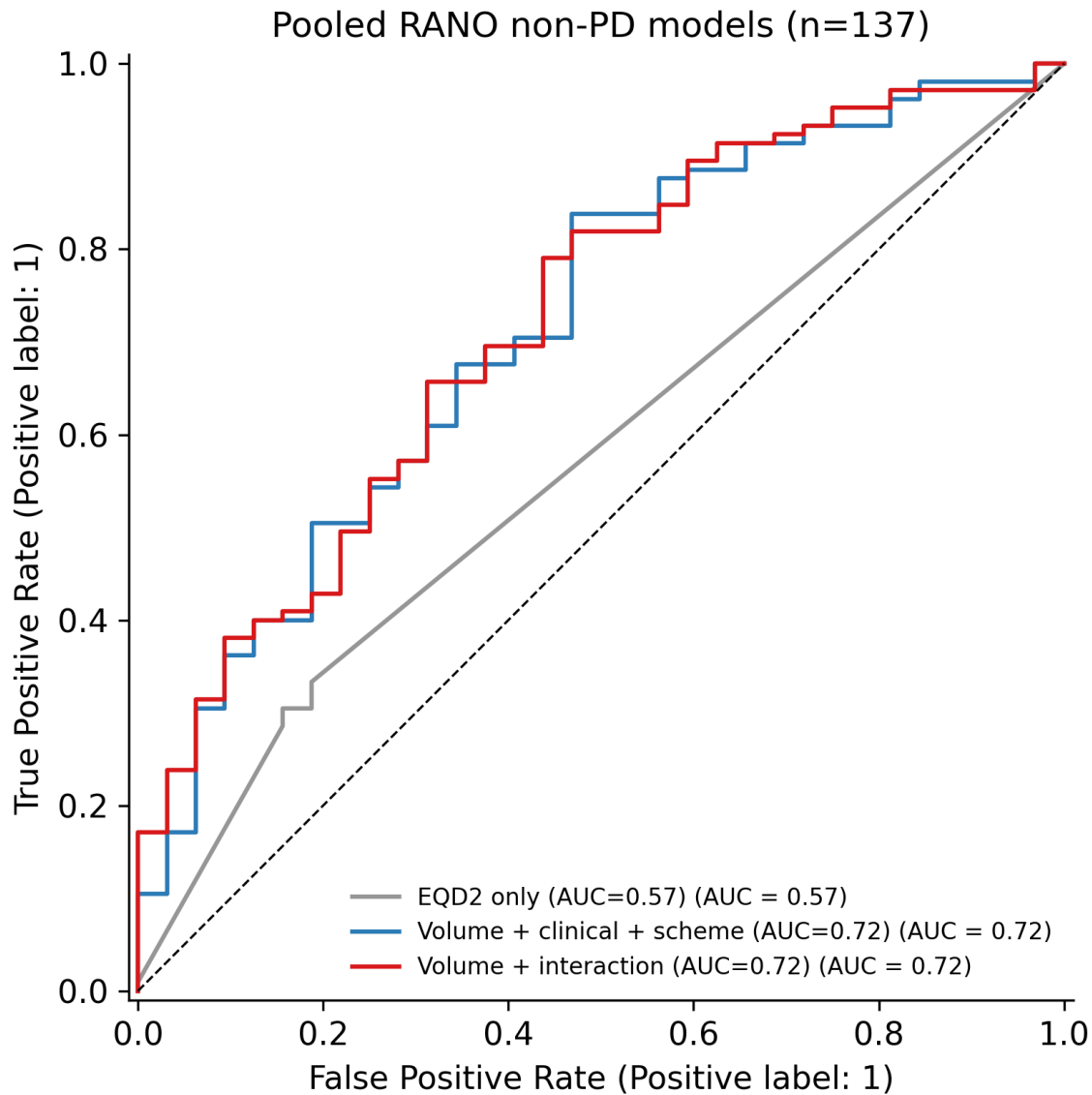


Figure 9: Figure 7. Pooled logistic ROC for RANO non-PD.

Figure 7. Pooled models on $n = 137$: EQD2 alone AUC approximately 0.57; GTV volume alone approximately 0.71; volume + age + WHO PS + scheme approximately 0.72.

Pooled model	n	AUC	LOOCV AUC
volume + age + PS + scheme	137	0.716	0.642
volume + age + PS + scheme + interaction	137	0.722	0.636

1.8.8 7.8 PyRadiomics vs DVH volume

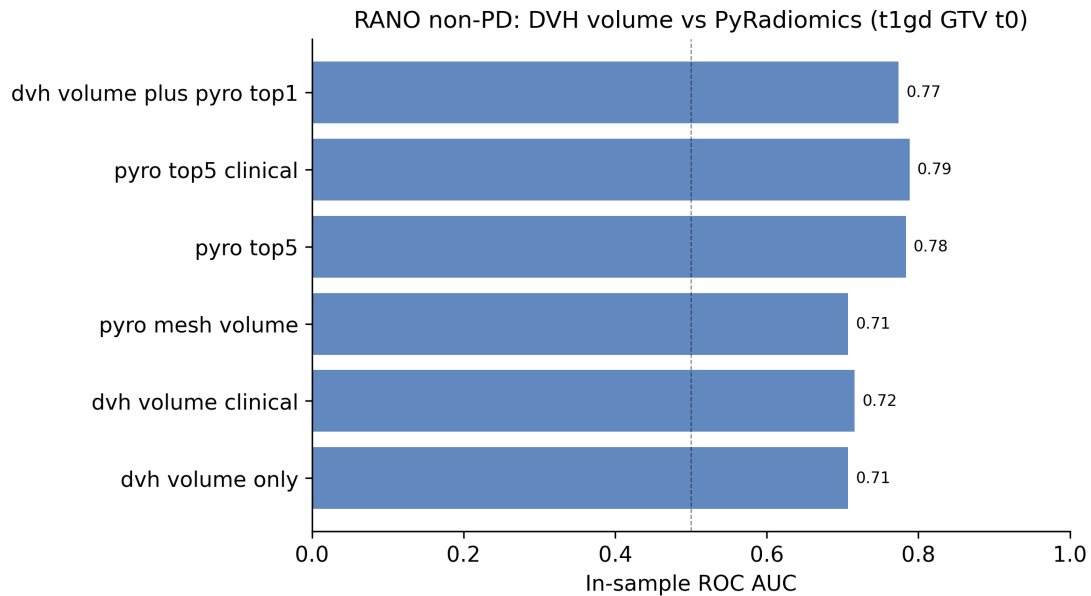


Figure 10: Figure 8. In-sample AUC: PyRadiomics top-5 vs DVH volume.

Figure 8. Author-provided PyRadiomics features (t1gd, GTV t0, Version 3 TSV) vs DVH GTV volume for RANO non-PD: top-5 radiomics AUC approximately 0.78 vs volume approximately 0.71 ($n = 137$).

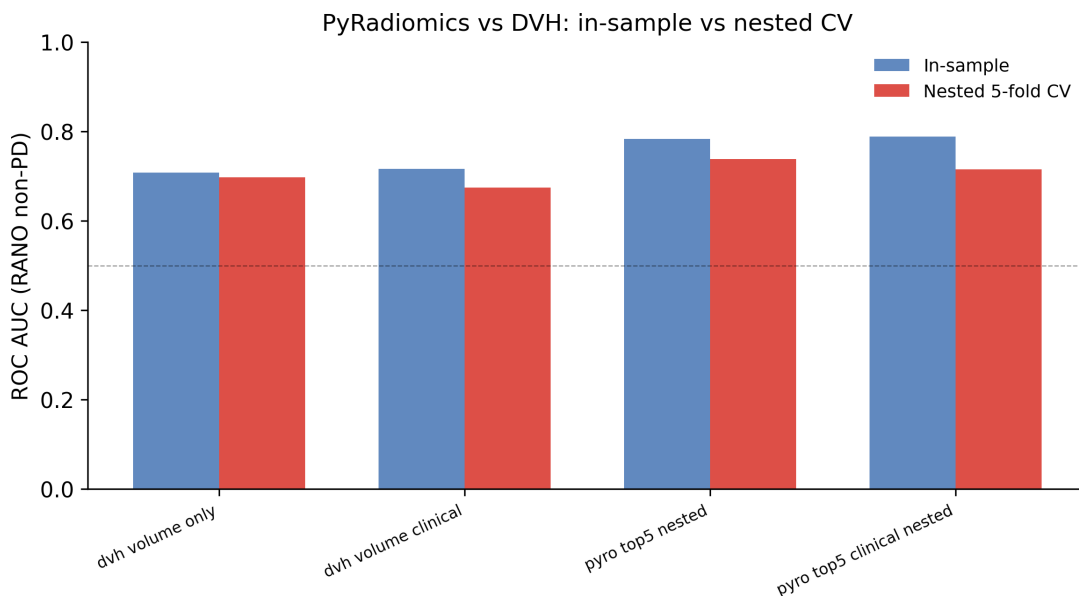


Figure 11: Figure 9. Nested cross-validation optimism check.

Figure 9. Nested 5-fold CV (top-5 feature selection on training folds only): radiomics nested AUC

approximately 0.74 vs volume approximately 0.70; optimism delta approximately 0.04–0.07.

Model (nested CV)	In-sample AUC	Nested CV AUC	Optimism delta
DVH volume only	0.707	0.697	0.010
PyRadiomics top-5	0.783	0.738	0.045
PyRadiomics top-5 + clinical	0.788	0.715	0.073

1.9 8. Survival Analysis

Kaplan-Meier and Cox proportional hazards models (lifelines) were fitted for overall survival.

1.9.1 8.1 Cox model with RANO (n = 137)

Covariate	Hazard ratio	p-value
RANO non-PD at t1	0.48	0.0009
WHO performance status	1.57	0.0005
EQD2 (per Gy)	0.97	0.009
Age (per year)	1.01	0.22

Concordance index: 0.667. RANO non-PD predicts longer OS independently of EQD2 and performance status, supporting the clinical relevance of the imaging endpoint even when dose-only TCP fails.

1.9.2 8.2 Clinical prognosis model (n = 190)

Covariate	Hazard ratio	p-value
60 Gy scheme (vs 40 Gy)	0.54	0.0007
WHO performance status	1.43	0.001
Age	1.01	0.23
Sex (male)	1.03	0.83

1.10 9. Literature Review and TCP Parameter Comparison

Literature sources are catalogued in `reports/literature_table.csv` (18 references). Link verification: `reports/literature_doi_check.md`.

1.10.1 9.1 Comparison of D50 and gamma50

Source	Model	Endpoint	n	D50 (Gy)	gamma50	Comparable to this study?
This study (CFB-GBM)	Poisson TCP (EQD2)	OS $\geq$ median (proxy)	190	53.2	3.32	reference
Maitre et al. 2020	TCP review	LC / NTCP (multiple)	review	40–80	1–5	partial
Okunieff et al. 1995	Poisson meta-analysis	Tumour control (mixed sites)	review	50–70	1–4	partial
Ohri et al. 2017	TCP/NTCP review	OS preferred	review	—	—	design guidance
Embring et al. 2020	DVH prognostic	OS (GBM)	120	—	—	endpoint differs
Gardner et al. 2024	Radiobiology review	Multi-scale	review	context	context	partial

Our bootstrap D50 (53 Gy) falls within published broad ranges but **must not be interpreted as a GBM tumour-control D50** because (i) the endpoint is OS proxy, not local control; (ii) dose spread within protocol is less than 1 Gy; (iii) two fractionation schemes are pooled.

1.10.2 9.2 Relation to CFB-GBM author radiomics work

Moreau et al. demonstrated imaging AI and PyRadiomics for treatment efficacy prediction on the same TCIA resource [9,10]. Our head-to-head comparison shows that author-provided t1gd GTV t0 radiomics modestly outperform scalar DVH volume for early RANO (nested CV AUC 0.74 vs 0.70), while DVH volume remains fully reproducible from RTDOSE and GTV without MRI preprocessing [17].

1.11 10. Discussion

1.11.1 10.1 Why pooled dose-TCP fails on CFB-GBM

CFB-GBM documents routine clinical practice, not a dose-escalation trial. Within the 60 Gy arm, every patient received essentially the planned protocol dose (GTV Dmean SD = 0.28 Gy). TCP

models mathematically require variation in the predictor; without it, fitted D50 and gamma50 reflect cohort mixing rather than a true dose-response law [7].

Using OS $\geq$ median as a TCP endpoint further confounds interpretation: higher EQD2 patients live longer primarily because they were younger and treated with curative-intent 60 Gy, not because incremental GTV dose within protocol controlled tumour. When RANO at t1 is used instead, pooled EQD2 alone performs worse than random (AUC approximately 0.43) because early imaging PD was more frequent in the 60 Gy arm despite better OS.

This negative result is scientifically valuable: it demonstrates that **endpoint and cohort design dominate model sophistication** in open-data TCP studies.

1.11.2 10.2 Where prognostic signal remains

When fractionation scheme and clinical covariates are included, **pre-treatment GTV volume** predicts RANO non-PD with moderate discrimination (pooled AUC approximately 0.72; LOOCV approximately 0.64). In the hypofractionated subgroup, the effect is stronger but the sample is small ($n = 34$; LOOCV AUC approximately 0.74 for volume only).

PyRadiomics features from the Version 3 TSV provide a modest incremental gain over volume (nested CV AUC approximately 0.74 vs 0.70). Optimism bias between in-sample and nested CV (delta approximately 0.04–0.07) underscores the need for pre-specified validation in future work.

1.11.3 10.3 Limitations

- No formal local control labels; RANO at t1 is an imaging surrogate, not pathological control [12,13].
- Single institution, retrospective, deceased-patients-only public release [8].
- Small hypofractionated RANO subset ($n = 34$); wide bootstrap CIs in 40 Gy arm.
- t1 GTV Nifti masks not validated locally; t1 volumes taken from author TSV.
- PyRadiomics features were author-provided; we did not re-extract locally (intentional parity with published baseline [9]).

1.11.4 10.4 Feasibility checklist for future open TCP studies

1. Report within-arm GTV Dmean IQR before any TCP fit [7].
2. Prefer local control or validated imaging response over OS alone [12].
3. Adjust for fractionation, age, and performance status [15].
4. Report LOOCV or nested CV alongside in-sample AUC [18].
5. Compare against author-provided radiomics baselines when available [8,9,17].

1.12 11. Conclusion

We implemented a reproducible TCP modeling workflow on the CFB-GBM cohort: Nifti-based DVH extraction (GTV only), four TCP model families, bootstrap and profile-likelihood uncertainty

intervals, model comparison (AIC, BIC, ROC, Brier), calibration assessment, survival analysis, literature comparison of D50 and gamma50, and RANO/PyRadiomics prediction with nested cross-validation.

Classical pooled TCP dose-response validation is not feasible on this routine-care dataset. With confounding addressed, **tumour burden metrics** (GTV volume and selected PyRadiomics features) predict early RANO with moderate out-of-sample discrimination.

1.13 References

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1.14 Appendix A — Reproducibility commands

```

pip install -r requirements.txt
make report
make verify-dois
make check-notebooks
make export-assignment

```

Plain-language glossary for group presentations: `reports/group_glossary_guide.md`

1.15 Appendix B — Repository outputs

Output	Path
Modeling table	data/processed/modeling_table.csv
Summary results	reports/RESULTS.md
This report (PDF)	reports/assignment_report.pdf

Output	Path
Scientific manuscript	reports/manuscript.pdf
Figure captions	reports/figure__captions.md
Literature table	reports/literature__table.csv